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"Comparative Analysis of Interpretable Machine Learning Models for Fetal Health Risk Classification Using CTG Data with SMOTE-Based Class Balancing"

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1.INTRODUCTION

Fetal mortality and perinatal morbidity represent a persistent global health crisis, disproportionately affecting low- and lower-middle-income countries (LMICs) where access to high-quality prenatal and intrapartum care remains critically limited. According to a joint report by the World Health Organization (WHO), UNICEF, and the World Bank, nearly two million stillbirths occur annually, equivalent to one every 16 seconds, with over 40% taking place during labor itself, the majority of which are considered preventable with timely clinical intervention [1]. LMICs account for 84% of all global stillbirths despite representing only 62% of all live births, with sub-Saharan Africa and Southern Asia collectively bearing three-quarters of this preventable burden [1]. A principal determinant of these avoidable perinatal losses is the failure to detect fetal distress—particularly hypoxia—in a timely and accurate manner [2].

Cardiotocography (CTG) has served as the cornerstone of intrapartum fetal surveillance for nearly six decades, simultaneously recording fetal heart rate (FHR) and uterine contractions (UC) to identify early indicators of fetal compromise. Despite its near-universal adoption in clinical obstetrics, the diagnostic utility of CTG is substantially limited by a relatively poor positive predictive value for abnormal traces and by significant inter- and intra-observer variability in interpretation [3]. Inter-observer reliability in CTG classification using established clinical guidelines, including FIGO, ACOG, and NICE, consistently yields Kappa values ranging from 0.3 to 0.6, reflecting only low to moderate agreement even among experienced practitioners [4]. Empirical evidence has consistently demonstrated poor predictive performance in manual CTG interpretation, with positive predictive values for adverse fetal outcomes remaining unacceptably low and clinical decision accuracy only marginally better than chance [3]. Beyond statistical limitations, manual interpretation is inherently constrained by human factors including cognitive

fatigue, subjective judgment, and variable clinician experience that collectively compromise diagnostic reliability and patient safety [5]. The clinical consequences of such errors are severe: False Negatives, in which fetal distress goes undetected, may result in delayed intervention, hypoxic-ischemic encephalopathy, and preventable perinatal death [6].

A. BACKGROUND

The rapid advancement of machine learning (ML) has introduced new opportunities for automated, objective, and scalable clinical decision support in obstetrics. By extracting complex, non-linear discriminative patterns from historical CTG recordings, ML classifiers can categorize fetal health status into three clinically defined classes—Normal, Suspect, and Pathological—with greater consistency and reproducibility than manual assessment. Scoping reviews have confirmed the broader applicability of supervised ML approaches to CTG-based fetal health classification across diverse clinical populations [3]. Rahmayanti et al. [7] reported classification accuracy between 89% and 99% using algorithms including K-Nearest Neighbors on the UCI CTG dataset, while Mehbodniya et al. [8] employed correlation-based feature analysis alongside multiple classifiers to examine the predictive contribution of individual CTG parameters to fetal health outcomes. Collectively, these findings position ML as a clinically promising complement to conventional CTG interpretation, particularly in settings where consistent and objective assessment is difficult to achieve.

B. PROBLEM STATEMENT

Despite these technological advances, a systematic appraisal of the existing literature reveals three persistent and clinically consequential methodological limitations. First, the predominant practice of optimizing and evaluating models solely on overall accuracy systematically obscures performance disparities across minority classes [6], [7]. In the clinical context of fetal health, a False Negative in the Suspect or Pathological category—wherein a compromised fetus is incorrectly classified as Normal—constitutes a missed diagnosis with potentially fatal consequences, including delayed intervention and irreversible neonatal injury. Second, the inherent class imbalance of the UCI CTG dataset—wherein Normal cases substantially outnumber

Suspect and Pathological instances—is frequently unaddressed or inadequately mitigated, systematically biasing classifiers toward majority-class predictions and artificially inflating reported accuracy [5], [9]. Third, studies employing ensemble or deep learning architectures, while achieving marginally superior aggregate accuracy, rely on "black-box" models that sacrifice the clinical interpretability required for physician trust and real-world deployment in critical care environments [9], [10].

C. RESEARCH OBJECTIVES

To directly address the identified gaps, this study proposes a rigorously designed comparative framework evaluating four interpretable, lightweight machine learning classifiers: K-Nearest Neighbors (KNN), Decision Tree (DT), Logistic Regression (LR), and Naive Bayes (NB). These four classifiers were deliberately selected on the basis of three clinically motivated criteria: inherent model transparency that enables direct decision traceability without post-hoc explanation, low computational overhead suitable for deployment in resource-constrained healthcare environments, and well-documented performance in structured tabular medical datasets [7], [8]. Unlike prior works that adopt overall accuracy as the primary evaluation criterion [6], [7], this study explicitly designates class-level Recall as the primary clinical metric, with the specific objective of minimizing life-threatening False Negatives in the Suspect and Pathological categories. Class imbalance is mitigated through the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic minority samples through interpolation rather than exact duplication—thereby preserving data diversity and reducing the overfitting risk associated with basic up-sampling or class weighting approaches [5]. SMOTE is applied exclusively within the training partition to prevent data leakage and preserve evaluation integrity. Model performance is further optimized through GridSearchCV hyperparameter tuning and validated via 10-fold stratified cross-validation, ensuring results that are both clinically meaningful and statistically sound.

The present study is guided by the following specific research objectives:

- **RO1:** To evaluate and compare the classification performance of four interpretable machine learning algorithms—KNN, DT, LR, and NB—on the UCI CTG dataset for three-class fetal health prediction.
- **RO2:** To investigate the effect of SMOTE-based class balancing on minority class Recall, specifically for the clinically critical Suspect and Pathological categories, with the objective of quantifying its impact on False Negative reduction.
- **RO3:** To identify the optimal interpretable classifier that maximizes class-level Recall while maintaining clinically acceptable overall accuracy and Macro F1-Score.
- **RO4:** To determine the most diagnostically significant CTG features through Decision Tree-based feature importance analysis, providing clinically interpretable insights for obstetric practitioners in resource-constrained environments.

The remainder of this paper is organized as follows: Section 2 provides a systematic review of related literature on ML-based fetal health classification and formally establishes the research gap. Section 3 details the dataset characteristics, preprocessing pipeline, and experimental methodology. Section 4 presents and critically discusses the comparative results with emphasis on clinical evaluation metrics. Section 5 concludes the study and proposes directions for future research.

2. LITERATURE REVIEW

The application of machine learning to automate CTG-based fetal health classification has attracted sustained and growing research interest over the past several years. A substantial body of work has demonstrated the potential of supervised learning algorithms to classify fetal health status from CTG-derived features. The following section provides a structured review of eight peer-reviewed studies that directly inform the motivation, methodological design, and evaluation framework of the present work. The reviewed studies are examined across four thematic areas, synthetic oversampling approaches, comparative algorithm evaluations, complex ensemble and deep learning architectures, and explainability-focused frameworks, with respect to their algorithmic approach, class imbalance handling, evaluation methodology, and key limitations—from which the research gaps addressed by the present work are formally derived.

Nazli et al. [2] conducted a comprehensive classification study on a CTG dataset comprising 2,126 patient records and 21 features, evaluating ten machine learning models including CatBoost, Decision Tree, ExtraTrees, Gradient Boosting, KNN, LightGBM, Random Forest, SVM, ANN, and DNN. To address class imbalance, SMOTE was applied during preprocessing. LightGBM achieved the highest performance with a classification accuracy of 90.73% and a balanced accuracy of 91.34%—the latter specifically noted for its effectiveness in handling imbalanced distributions. While this study represents one of the most comprehensive recent evaluations on the CTG dataset and correctly applies SMOTE, the study does not provide isolated class-level Recall analysis for the Suspect and Pathological categories. The inclusion of complex architectures such as DNN limits clinical interpretability, and no systematic hyperparameter optimization via GridSearchCV was reported, which constrains the reproducibility and optimization of the comparative findings.

Das et al. [5] proposed a soft-computing framework applied separately to both stages of labor—an important methodological distinction, as the two stages produce markedly different fetal heart rate dynamics. The study noted that precise interpretation of suspected fetal health cases remains particularly challenging, and that standard classification approaches often fail to adequately capture the distinct CTG patterns characterizing each labor stage. SMOTE was applied exclusively to the training data to preserve the integrity of the test distribution, thereby avoiding data leakage—

a methodologically rigorous practice that many studies neglect. The proposed model achieved strong AUC-ROC values across both labor stages. However, the study evaluated complex classifiers including SVM, Random Forest, MLP, and Bagging, without including interpretable lightweight algorithms such as Logistic Regression, Naive Bayes, or standalone Decision Trees. Class-level Recall for minority categories was not explicitly reported, and the clinical implications of False Negatives in Suspect cases were not discussed.

Rahmayanti et al. [7] conducted a comparative evaluation of seven classifiers—ANN, LSTM, XGBoost, SVM, KNN, LightGBM, and Random Forest—on the UCI CTG dataset of 2,126 records, reduced to 1,944 after outlier removal. Parameter optimization was conducted via grid search cross-validation, and class-level Precision, Recall, and F1-Score were reported for all three categories. Five of seven algorithms achieved accuracy between 89% and 99%, with LightGBM performing consistently across all evaluation scenarios. However, the study relied on basic up-sampling rather than SMOTE, and focused exclusively on complex architectures, entirely omitting interpretable lightweight classifiers such as standalone Decision Tree, Logistic Regression, and Naive Bayes.

Mushtaq and Veningston [6] compared six baseline classifiers, Logistic Regression, KNN, Naive Bayes, SVM, Random Forest, and Gradient Boosting, alongside a deep learning model that achieved 99% accuracy and 93% sensitivity, outperforming all baselines. This study provides direct performance benchmarks for three of the four algorithms evaluated in the present work. However, SMOTE was not employed, class-level Recall for the Suspect and Pathological categories was not individually reported, and the clinical tradeoff between Precision and Recall for minority classes was not systematically examined. Despite the use of post-hoc SHAP-based explainability, the inherent transparency of standalone lightweight classifiers, which require no post-hoc methods, remains preferable for resource-constrained clinical environments.

Mehbodniya et al. [8] deployed SVM, Random Forest, MLP, and KNN on CTG data alongside regression and correlation analysis to examine individual feature contributions. Random Forest achieved the best performance with 94.5% accuracy, 94% Recall, and 93% F1-Score through 10-fold cross-validation. However, Logistic Regression, Naive Bayes, and standalone Decision Tree

were excluded, SMOTE was absent from the pipeline, and class-level Recall for the minority Suspect and Pathological categories was not independently isolated—a significant limitation given that high overall accuracy can mask clinically dangerous False Negative rates.

Kuzu and Santur [10] proposed an ensemble framework for three-class fetal health classification using CTG data from nonstress tests, achieving accuracy above 99.5%, one of the highest reported values in the literature. Despite this performance, the ensemble architecture introduces considerable model complexity that limits clinical interpretability. The study did not evaluate interpretable standalone classifiers, and the extremely high accuracy raises concerns about potential overfitting within the complex stacked ensemble architecture. Class-level performance metrics, particularly Recall for the Suspect and Pathological categories, were not individually analyzed. Furthermore, while the authors utilized adaptive class weighting, they did not employ any robust synthetic oversampling technique like SMOTE.

Sirisha et al. [9] proposed a Dynamic Multi-Layer Perceptron (DMLP) evaluated progressively from single- to multi-layer configurations, applying data balancing, feature scaling, normalization, and hyperparameter tuning alongside an ablation study to quantify preprocessing contributions. The DMLP achieved 97% accuracy, outperforming traditional baselines including Decision Tree, Logistic Regression, and KNN. While this study demonstrates the value of systematic preprocessing—methodological practices that inform the present work—the deep neural architecture lacks the inherent clinical interpretability required for physician-assisted decision making, and class-level Recall for minority categories was not isolated.

Yin and Bingi [11] introduced a RISE-based feature importance framework combining SVM with individual CTG feature explainability analysis, identifying ML transparency as essential in resource-limited obstetric settings. While valuable for explainability, SVM offers limited inherent interpretability relative to Decision Trees or Logistic Regression, the four classifiers central to the present work were not evaluated, and class-level Recall was not addressed as a primary objective.

Across the reviewed studies, a consistent pattern emerges: while preprocessing rigor and algorithmic complexity vary, systematic evaluation of class-level Recall for clinically critical

minority categories remains absent, and interpretable lightweight classifiers are consistently overlooked in favor of complex architectures.

2.1 SUMMARY OF RESEARCH GAPS

A structured synthesis of the reviewed literature reveals three persistent and clinically consequential gaps that directly motivate the present study.

Gap 1 — Overreliance on Overall Accuracy as the Primary Metric. Across the reviewed studies, overall accuracy is consistently used as the primary, and frequently sole—evaluation criterion [7], [8], [9]. While accuracy provides a useful general performance measure, it systematically obscures minority class performance in imbalanced datasets. In the clinical context of fetal health, a False Negative in the Suspect or Pathological category, wherein a compromised fetus is classified as Normal, may result in delayed intervention, hypoxic-ischemic encephalopathy, and preventable perinatal death. No reviewed study explicitly designated class-level Recall as the primary evaluation metric with the specific clinical objective of minimizing life-threatening False Negatives.

Gap 2 — Inadequate Treatment of Class Imbalance and Its Impact on Minority Class Detection. Although several studies acknowledged the class imbalance inherent in the UCI CTG dataset, the specific impact of imbalance correction on minority class Recall was rarely isolated or reported independently. While Das et al. [5] correctly applied SMOTE exclusively to training data, the effect of this correction on Suspect and Pathological class Recall was not reported independently. Similarly, Rahmayanti et al. [7] applied basic up-sampling and Kuzu and Santur [10] utilized adaptive class weighting; however, these approaches lack the synthetic generation capability of SMOTE, which prevents information loss and reduces overfitting risks associated with exact minority sample duplication. The remaining reviewed studies, including Mushtaq and Veningston [6] and Mehbodniya et al. [8], neglected class imbalance entirely, resulting in classifiers systematically biased toward the majority Normal class.

Gap 3 — Insufficient Focus on Clinical Interpretability. The majority of high-performing approaches rely on ensemble methods [10] or deep learning architectures [6], [9] that function as computationally complex systems incompatible with the transparency requirements of clinical

decision support, where physicians must understand, audit, and trust model predictions. A systematic comparison of inherently interpretable, lightweight classifiers with explicit evaluation of minority class performance remains notably absent from the literature.

The present study directly addresses all three identified gaps by evaluating four interpretable classifiers, KNN, DT, LR, and NB, with class-level Recall as the primary metric, SMOTE applied exclusively to training data, GridSearchCV hyperparameter optimization, and 10-fold stratified cross-validation, providing a clinically grounded and methodologically rigorous comparative framework.

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